

03. Operators & Variables Quiz

1. What does the `//` operator do?

- A) Regular division
- B) Floor division
- C) Exponentiation
- D) Modulo

Explanation: `//` returns the largest integer less than or equal to the division result (e.g., `7 // 2 = 3`).

2. Which operator has the highest precedence?

- A) `+` (addition)
- B) `*` (multiplication)
- C) `()` (parentheses)
- D) `=` (assignment)

Explanation: Parentheses have the highest precedence in Python.

3. What is the output of `3 ** 2`?

- A) 6
- B) 9
- C) 1.5
- D) Error

Explanation: `**` is the exponentiation operator ($3^2 = 9$).

4. Which is a valid variable name?

- A) `2var`
- B) `_var2`
- C) `break`
- D) `var-name`

Explanation: Variable names cannot start with numbers, use hyphens, or be keywords.

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5. What does `%` operator do?

- A) Divides numbers
- B) Returns remainder
- C) Multiplies numbers
- D) Rounds numbers

Explanation: `%` is the modulo operator (e.g., $7 \% 3 = 1$).

6. Which statement assigns multiple variables?

- A) `x = 1; y = 2`
- B) `x, y = 1, 2`
- C) `x = y = 1`
- D) All of the above

Explanation: All are valid, but B demonstrates tuple unpacking.

7. What is the output of `not(5 > 2)`?

- A) `True`
- B) `False`
- C) `5`
- D) `2`

Explanation: `5 > 2` is `True`, so `not(True)` returns `False`.

8. Which operator checks identity?

- A) `==`
- B) `is`
- C) `=`
- D) `in`

Explanation: `is` checks if two objects are the same in memory.

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9. What does ``+=`` do?

- A) Adds 1 to a variable
- B) Adds a value to a variable
- C) Compares values
- D) Multiplies values

Explanation: ``x += 3`` is shorthand for ``x = x + 3``.

10. Which is an invalid assignment?

- A) ``a = b = 0``
- B) ``a, b = 1, 2``
- C) ``3 = x``
- D) ``x = y = z = 1``

Explanation: Left-hand side of ``=`` must be a variable, not a literal.

11. What is the output of ``True + False``?

- A) ``0``
- B) ``1``
- C) ``True``
- D) ``Error``

Explanation: ``True`` is ``1``, ``False`` is ``0`` (`1 + 0 = 1`).

12. Which operator checks membership?

- A) ``==``
- B) ``is``
- C) ``in``
- D) ``&``

Explanation: ``in`` checks if an item exists in a sequence.

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13. What does `~5` return?

A) `-5`

B) `-6`

C) `5`

D) `4`

Explanation: `~` is bitwise NOT (inverts bits: `~5 = -6`).

14. Which has the lowest precedence?

A) `*`

B) `**`

C) `+`

D) `=`

Explanation: Assignment operators have the lowest precedence.

15. What is the output of `bool("")`?

A) `True`

B) `False`

C) `None`

D) `""`

Explanation: Empty strings are falsy in Python.

16. Which is a ternary operator?

A) `if-else`

B) `x if condition else y`

C) `switch-case`

D) `for-else`

Explanation: Ternary operators condense if-else into one line.

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17. What does `<<` do?

- A) Right shift
- B) Left shift
- C) Logical AND
- D) Exponentiation

Explanation: `<<` shifts bits left ($5 << 1 = 10$).

18. Which variable name follows PEP 8 conventions?

- A) `myVariable`
- B) `my\_variable`
- C) `MyVariable`
- D) `mYvArIaBlE`

Explanation: PEP 8 recommends snake_case for variables.

19. What is the output of `0 and 5`?

- A) `0`
- B) `5`
- C) `False`
- D) `True`

Explanation: `and` returns the first falsy value (0).

20. Which operator is overloaded for string concatenation?

- A) `+`
- B) `\*`
- C) `&`
- D) `||`

Explanation: `+` concatenates strings ("a" + "b" = "ab").